Ratios and Equivalent Ratio Tables

Reading ratios off models, part-to-part and part-to-whole forms, equivalent ratio tables, and the value of a ratio

Directions. Every problem hands you a model — a sticker strip, a tape diagram, or a ratio table — and asks you to turn it into notation. Remember the three written forms of the same comparison:

- words: a **to** b colon: $a : b$ fraction: $\frac{a}{b}$
- the quantity named *first* in the question is written first

Show the model work you used, then write the answer on the line.

1. Sticker strip. The strip below is one sheet of stickers. Each S is a star sticker and each M is a moon sticker.

S	S	S	S	S	S	M	M	M	M	M	M	M	M

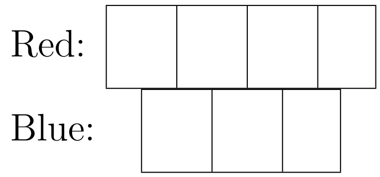
Write the ratio of **star stickers to moon stickers** in all three forms, then give that ratio in simplest form as a fraction.

Answer: __________________

2. Same sticker strip. This time compare a part with the whole sheet. Write the ratio of **star stickers to all the stickers** as a fraction in simplest form, and say in words what that fraction tells you about the sheet.

Answer: __________________

3. Tape diagram. A bracelet uses red beads and blue beads in the ratio 4 to 3. The diagram shows one box for each part of the ratio, and every box holds the same number of beads.



Each box holds 6 beads. How many beads are on the whole bracelet?

Answer: ___________________

4. Ratio table. A bead kit is packed so that there are 4 red beads for every 6 white beads.

White beads	6	3	15
Red beads	4		

Fill in both empty cells. Say which way you scaled to reach each one.

Answer: ___________________

5. Value of a ratio. A bottling machine fills 84 bottles in 4 minutes, at a steady rate. Write the ratio of bottles to minutes, then find its **value**: the number of bottles filled in one single minute.

Answer: ________________

6. Part to whole. A shelf holds only nonfiction books and fiction books, in the ratio $5 : 3$. What **fraction of all the books on the shelf** is fiction? Explain where the denominator of your fraction comes from, since it is not one of the two numbers in the ratio.

Answer: ________________

7. Ratio table. Concert tickets are sold at one steady price: 15 tickets cost 12 dollars.

Tickets	15	5	25
Cost (dollars)	12		

Complete the table. What does a block of 25 tickets cost, in dollars?

Answer: ________________

8. Comparing by value. Two drink mixes are made from scoops of powder and liters of water.

Mix A: 3 scoops in 5 liters. Mix B: 5 scoops in 8 liters.

Write each mix as a ratio in fraction form, find the value of each ratio, then write $<$ or $>$ between them in this order: value of Mix A ______ value of Mix B.

Answer: ________________

9. Ratio table, harder step. A lemonade recipe uses 12 lemons for every 8 cups of water.

Cups of water	8	2	30
Lemons	12		

Fill in the middle column first — dividing is the easy step — then use that column to reach 30 cups. How many lemons does the 30-cup batch need?

Answer: ________________

10. Synthesis. A mural paint is mixed from blue paint and yellow paint in the ratio 5 to 4. Ravi needs 63 liters of mixed paint in total.

Draw a tape diagram with one box per share, work out how many liters one box is worth, then find how many liters of **blue** paint Ravi needs. Write the ratio of blue paint to total paint as a fraction as part of your reasoning.

Answer: ________________ L